

Animal Implicatures: A Note on the Informativity Principle and Bayesianism*

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Abstract

Several recent analyses of animal signals have proposed that when animals have the choice between a signal S and a more informative signal S^+ , S^+ is preferred if it is true (= Informativity Principle). The principle is usually taken to apply both in production and in comprehension. But some uses of the principle in comprehension can be replaced with simple Bayesian principles. We explain in which cases this happens, and why the key data adduced for the Informativity Principle cannot be explained in this way because they involve production rather than comprehension. Thus simple Bayesianism would need to be supplemented with additional principles to derive the Informativity Principle.

Keywords: animal linguistics, animal communication, animal semantics, animal signals, Bayesian pragmatics, Informativity Principle

1 The Informativity Principle

Several recent analyses of animal signals developed within “animal linguistics” (e.g. Schlenker et al. 2014, 2016a,b,c) have postulated an Informativity Principle:

(1) Informativity Principle

Assume that the semantics yields a relation ‘is strictly more informative than’ on some signals that are alternatives to each other. Underinformative signals are prohibited by the following rules:

- a. *Production*: Do not utter S in a situation w if a strictly more informative alternative S^+ is true in w .
- b. *Comprehension*: If you perceive S in a situation w , infer that every strictly more informative alternative S^+ is false in w .

(from Schlenker et al. 2016c)

In its human incarnation, the Informativity Principle derives of so-called “implicatures”: an expression S that competes with a more informative expression S^+ is pragmatically enriched to S and not S^+ (e.g. Grice 1989; Goodman and Frank 2016; Franke and Jäger 2016; Horn 1972).

A key motivation for the animal version of the Informativity Principle comes from the stylized observation

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in (2a).¹

(2) Stylized motivation for the Informativity Principle

Several species have a call S used for diverse dangerous situations (e.g. situations where there is a predation and non-predation-related situations), and a call S^+ used only for specific predators, typically eagles. Then:

- a. *Production* [actual argument]: When there is an eagle stimulus, signalers standardly produce sequences of S^+ signals, not of S signals.
Examples: Titi monkeys (Schlenker et al. 2016b), Putty-nosed monkeys (Schlenker et al. 2016a)
- b. *Comprehension* [possible argument]: When there is an S signal, receivers behave as if there were a non-eagle-related threat.

There is considerable evidence that animals perform Bayesian reasoning, which is essential to fitness (see for instance Valone 2024 for a recent review). It is thus natural to ask whether Bayesian principles could obviate the need for the Informativity Principle. We will argue that the answer is positive for the Informativity Principle in comprehension (= (1b)) but not for the Informativity Principle in production (= (1a)). Since some crucial data for the Informativity Principle come from production (= (2a)) rather than comprehension (= (2b)), simple Bayesian principles do not suffice to eliminate the Informativity Principle. Simple Bayesianism would thus need to be supplemented with additional principles to derive the Informativity Principle in production.

A disclaimer before we start. For an argument for the Informativity Principle as stated in (1) to get off the ground, one signal must be strictly more informative than another signal. But it is hard to find incontrovertible evidence for this situation. For instance, in Titi monkeys the A-call has been analyzed as raising a serious non-ground threat, while the B-call has been taken to be used for any threat (Schlenker et al. 2016b). The effect of the Informativity Principle could be mimicked without it by just positing that the B-call means: non-[serious non-ground] threat. This would trivialize the entire discussion that follows, as call meanings without additional principles would suffice to derive the observed behaviors. One can take this alternative B-meaning to be unnatural, but it is hard to rule it out. Furthermore, Steinert-Threlkeld et al. 2021 show that under certain evolutionary assumptions, general calls might end up acquiring a specialized meaning of precisely this kind. To ensure that general calls are stable despite this result, one might for instance have to posit that some of the undesirable specialized meanings are cognitively impossible. We have nothing to contribute to this discussion in the present piece, and in the remainder we assume that general calls are real.

2 A simple Bayesian alternative in comprehension

Let us consider again a Titi monkey that hears a B-call, and let us assume that there is no Informativity Principle whatsoever, be it in production or in comprehension. Let us further assume that a signaling Titi has the following behavior:

- If there is an eagle, use A with probability .5 and B with probability .5
- If there is a non-eagle-related threat, use B with probability 1.

Now suppose that relative to all threats, eagle-related threats occur with probability .5, while non-eagle-related threats occur with probability .5.

These assumptions are summarized in (3).

(3)

a. Situations:	Eagle ($p = .5$)	Non-eagle ($p = .5$)
b. Calls:	A ($p = .25$) B ($p = .25$)	B ($p = .5$)

It is clear that a Titi receiver that hears a B-call will take it to be more likely that there is a non-eagle rather than eagle threat, since conditional on hearing B, the probability of encountering a non-eagle is twice that

¹Throughout, we use the term "threat" broadly, as a general alarm call could be used for non-threatening but noteworthy events.

of encountering an eagle, as is graphically shown in (3b) (the probability of ‘Eagle & B’ is 0.25, while that of ‘Non-eagle & B’ is 0.5). More precisely, $P(\text{Non-eagle}|B) = 2/3$.² But this means that the effect of the Informativity Principle in perception (= (1b)) is emulated without any Informativity Principle.

It is clear that this conclusion does not necessarily follow. If non-eagle threats have probability .1 rather than probability .5, as illustrated in (4), while eagle-related threats give rise to half of A-behavior and half of B-behavior, as before, the reasonable reaction when hearing a B-call is to think that there is probably an eagle, not that there is a non-eagle, since we now have $P(\text{Non-eagle}|B) \simeq 0.18$.³

(4)

a. Situations:	Eagle ($p = .9$)	Non-eagle ($p = .1$)
b. Calls:	A ($p = .45$)	B ($p = .45$)
		B ($p = .1$)

Let us now consider things in a slightly more general case, as described in (5).

(5) **General case**

a. *Situations*

Among all threats, eagle threats occur with probability e , non-eagle threats with probability $(1 - e)$, with $0 < e < 1$:

$$P(\text{Eagle}) = e, \quad P(\text{Non-eagle}) = 1 - e$$

b. *Calling strategies*

Eagle threat situation: use A with probability $1 - b$, B with probability b , with $0 < b < 1$:

$$P(A | \text{Eagle}) = 1 - b; \quad P(B | \text{Eagle}) = b$$

Non-eagle threat situation: use B with probability 1

$$P(A | \text{Non-eagle}) = 0; \quad P(B | \text{Non-eagle}) = 1$$

Now we can use Bayes’s rule to compute the conditional probability of being faced with an eagle vs. a non-eagle threat contingent on hearing A or B.

(6) **Bayes’s rule**

$$P(X | Y) = \frac{P(Y | X) P(X)}{P(Y | X) P(X) + P(Y | \neg X) P(\neg X)} \cdot P(X)$$

²By inspecting the table in (3), one sees that $P(\text{Non-eagle}|B) = \frac{P(\text{Non-eagle} \& B)}{P(B)} = \frac{0.50}{0.75} = 2/3$

³By inspecting the table in (4), one sees that $P(\text{Non-eagle}|B) = \frac{P(\text{Non-eagle} \& B)}{P(B)} = \frac{0.1}{0.55} \approx 0.18$

(7) Application

a.

$$\begin{aligned}
 P(\text{Eagle} \mid B) &= \frac{P(B \mid \text{Eagle})}{P(B \mid \text{Eagle})P(\text{Eagle}) + P(B \mid \text{Non-eagle})P(\text{Non-eagle})} \cdot P(\text{Eagle}) \\
 &= \frac{be}{be + 1 \cdot (1 - e)} \\
 &= \frac{be}{be + (1 - e)} \quad \left(= \frac{1}{1 + \frac{1 - e}{be}} \right)
 \end{aligned}$$

b.

$$P(\text{Non-eagle} \mid B) = 1 - \frac{be}{be + (1 - e)} = \frac{1 - e}{be + (1 - e)} \quad \left(= \frac{1}{1 + \frac{be}{1 - e}} \right)$$

Upon hearing an A-call, the receiver can infer that there is an eagle, as expected.⁴ When the receiver hears a B-call, the situation is more complex, as shown by (7). In all cases, after hearing a B-call, the hearer increases its assessment of the probability of non-Eagle threat, relative to an Eagle-threat, as shown in (8).⁵

(8) $P(\text{Non-eagle} \mid B) > P(\text{Non-eagle})$

Proof: $be < e$ (since $0 < b < 1$), therefore $be + (1 - e) < e + (1 - e) = 1$, therefore $(1 - e)/(be + (1 - e)) > (1 - e)$. $\square$

To illustrate, in (4) the prior probability of a Non-eagle was very low (0.1), but the posterior probability of a Non-eagle given after hearing a B-call is approximately .18,⁶ which is still low, but nearly twice as much as the prior probability.

As a result, whatever the value of e and b , upon hearing a B-call, a Bayesian listener should pay more attention than before to the possibility of non-eagle-related threats. This is true even in the absence of any Informativity Principle, and even if b is high, i.e. even if the B-call (i.e. the general call) tends to be favored across the board, in violation of the Informativity Principle. In particular, if eagle-related threats are no more frequent than non-eagle-related ones ($e \leq 0.5$), as we assumed in (3), then whatever the value of b is, $P(\text{Non-eagle} \mid B) \geq 0.5$.⁷ That is, in a situation where initially non-eagle-related threats and eagle-related threats are equally likely, B will lead to an inference that a non-eagle-related threat is likely as soon as the A-call is sometimes used for eagles, even if in general B-calls are preferred.

More generally, the formula in (7b) implies that the posterior probability of a non-eagle-related threat after hearing a B-call increases when e and b decrease. That is, the function

$$f(e, b) = \frac{1}{1 + \frac{be}{1 - e}} = P(\text{Non-eagle} \mid B)$$

is decreasing with respect to its two arguments.⁸ Intuitively, this is because when the prior probability of an eagle-related-threat (e) is low, its posterior probability will tend to be low as well, and if the probability (=

⁴The formula reduces to $P(\text{Eagle} \mid A) = (1 \cdot e)/(e + 0) = 1$.

⁵All the computations provided here are relative to a situation in which it is already known that there is either an eagle or a non-eagle-related threat. When hearing a B-call, the receiver learns that there is a threat, so typically both eagle and non-eagle threats become more probable than before, and if eagle-related threats are antecedently much more likely than non-eagle-related ones, the receiver should be more concerned about eagles than non-eagles after hearing a B-call. The point we make here is that a B-call increases the probability share of non-eagle-related threats relative to that of eagle-related threats.

⁶To reiterate footnote 3, $P(\text{Non-eagle} \mid B) = .1/.55 \approx .18$. This corresponds to (7b) with $e = .9$ and $b = .5$, hence $P(\text{Non-eagle} \mid B) = .1/(.45 + .1)$.

⁷If $e \leq 0.5$, $be/(1 - e) < 0.5 \times b/(1 - 0.5) = b$, hence $P(\text{Non-eagle} \mid B) = 1/(1 + be/(1 - e)) > (1/(1 + b)) > 0.5$ (since $b < 1$).

⁸Proof: be is increasing in e , and so is $1/(1 - e)$, and therefore $eb/(1 - e)$ is increasing in e . ($be/(1 - e)$) is also of course increasing in b . So $[1 + be/(1 - e)]$ is increasing in both e and b , and therefore $1/[1 + be/(1 - e)]$ is decreasing in both e and b .

Now in simple cases we can assume that the receiver will adopt a non-eagle-related behavior just in case the probability of eagle presence after hearing B is less than a certain threshold θ , as stated in (9).⁹

$$P(\text{Eagle} \mid B) < \theta$$
$$\frac{be}{be + (1 - e)} < \theta$$
$$(1 - \theta) be < \theta(1 - e)$$
$$(10) \quad be < 1 - e \iff e(1 + b) < 1$$

For simplicity, we have stated the predictions in terms of eagle-related vs. non-eagle-related threats, although analyses typically posit slightly different meanings. For instance, for Titi monkeys, the A-call has been treated as raising a “serious non-ground” threat (Schlenker et al. 2016b)—or sometimes as “serious” threat, with the assumption that serious threats are all non-ground (Commier and Berthet 2019). The B-call has been treated as a general threat call. Cäsar 2012 assessed in detail Titi gaze orientation after hearing A- vs. B-calls. It is clear that receivers expected a non-ground threat after hearing A-calls, as such playbacks “elicited longer looking in the upwards direction, indicating monkeys were anticipating an important event, such as a raptor attack, from above” (Cäsar 2012 p. 165). For B-calls the situation is a bit less clear: “Playback elicited longer looking towards the caller, indicating monkeys were anticipating an attack/event from near the caller”, possibly because “listeners were looking for cues from caller’s behaviour and/or body orientation” (p. 166).

Crucially, the Titi monkey data that were analyzed using the Informativity Principle involved production. To illustrate, we have copied in (11) the results of Căsar’s experiment involving a model raptor in the canopy, with each line representing the first 30 calls produced. Virtually all of them are A-calls, there are a few O-calls (of which we know nothing), and no B-calls at all.

[illegible]

Similar facts hold in Putty-nosed monkeys, which also have a general-like ‘pyow’ call and an eagle-related

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‘hack’ call. In experiments involving eagle shrieks or eagle visual models, Arnold et al. (Arnold et al. 2008) found that overwhelmingly, sequences produced started with hack, not pyow, as seen in (12).

(12) Arnold et al.’s experiments with eagle shrieks and eagle visual models (Arnold et al. 2008)

Eagle acoustic model

No.	N	1	2	3	4	5	6	7	8	9	10	11
1	23	H	H	H	H	P	H	H	P	P	P	P
2	8	H	H	H	H	H	H	H				
3	29	H	H	H	H	H	P	H	H	H	H	H
4	13	H	H	H	H	H	P	P	P	P	P	P
5	38	H	H	H	H	H	H	H	H	H	H	H
6	33	H	H	P	H	P	H	H	H	H	H	H
7	12	H	H	H	H	H	H	H	H	H	H	H
8	12	P	P	P	P	P	P	P	P	P	P	P
9	40	H	H	H	H	H	H	H	H	H	H	H
10	13	H	H	H	H	H	H	H	H	H	H	H
11	20	H	H	H	H	H	H	H	H	H	H	H
12	5	H	H	H	H	H						
13	14	H	H	H	H	H	H	H	P	P	P	P
14	19	H	H	H	H	H	H	H	H	P	P	P
15	13	H	H	H	P	P	P	P	P	P	P	P
16	19	P	P	P	P	P	P	P	P	P	H	P
17	12	H	H	H	H	P	P	P	P	P	P	P
18	7	H	H	H	H	H	H	H				
19	2	H	H									
20	17	H	H	H	H	H	H	H	H	H	H	H

Eagle visual model

N	1	2	3	4	5	6	7	8	9	10	11
11	H	H	H	H	P	P	P	P	P	P	P
33	P	P	P	P	P	P	P	P	P	P	P
6	H	H	H	H	P	P					
8	H	H	H	H	P	P	P	P			
3	H	H	H								
57	H	H	H	H	H	H	H	H	H	H	H
5	H	H	H	H	H						
10	H	H	H	H	P	H	P	P	P	H	

Each line represents the series of (the first 11) Putty-nosed calls given in response to visual and acoustic models of a crowned eagle (hack-calls (H), pyow-calls (P))

The Informativity Principle in production captures these data. Simple Bayesianism does not: it is an inferential mechanism, which has nothing to say about the choice between two signals in production. Thus to derive the desired data, simple Bayesianism must be supplemented with additional principles.

It should be noted that in some cases, eagle calls are followed by general calls. This pattern is illustrated in Putty-nosed sequences in (12), and it is also found in other Titi experiments (Căsar 2012; Căsar et al. 2013; Schlenker et al. 2016b). A standard line of analysis is that in some cases, the level of threat typically decays after some time (Schlenker et al. 2016a,c). If Putty-nosed hack has a ‘serious non-ground’ component, alarm decay has the consequence that after a while hack can’t be used (there is a non-ground threat, but a weak one), and thus only pyow can be used.¹⁰

A separate point is that once the Informativity Principle is present in production, Bayesian reasoning

¹⁰Two points should be added. First, in Putty-nosed monkeys, a more sophisticated meaning was explored for hack, namely ‘serious non-ground movement-related threat’. The ‘movement-related’ part was intended to capture the meaning of so-called ‘pyow-hack sequences’, which are correlated with group movement. Second, in Titi monkeys, pure A-sequences were found in ‘model raptor in the canopy’ situations while A-sequences followed by B-sequences were found in ‘model raptor on the ground’ situations. For the latter observation, it was postulated (Schlenker et al. 2016b) that “a raptor on the ground will attack (if it does) by flying”, hence explaining the initial A-component; but that “immobility on the ground is not a typical hunting position”, which explains why “after a while the threat stops being considered as serious—presumably because a raptor would not normally remain motionless on the ground for long periods of time”.

obviates the need to posit it as a primitive in comprehension.¹¹ The reason is that $b = P(B \mid \text{Eagle}) = 0$, which essentially guarantees that the inequality in (9) is satisfied as soon as there are non-eagle situations (i.e. $e \neq 1$).¹²

4 Conclusion

In sum, while simple Bayesianism could in principle obviate the need for some uses of the Informativity Principle applied in comprehension, this is not so for the Informativity Principle applied in production. Crucially, some of the most striking arguments for the Informativity Principle come from production data—hence the latter is not dispensable as things stand. Simple Bayesianism would need to be supplemented with further principles to derive these data.¹³ Of course, one could in principle posit that the Informativity Principle is applied in production while simple Bayesianism is applied in comprehension. But this would still grant a major component of the analysis that Bayesianism was intended to replace.

These remarks have consequences for the evolutionary emergence of the Informativity Principle. In Schlenker et al. (forthcoming), it was shown that there is a clear evolutionary path for the principle if it emerges in production first, and in comprehension second (essentially because, on such a scenario, each evolutionary step only produces better information and thus greater fitness).¹⁴ If Bayesian reasoning is universally available, we might dispense with the second step, as the effect of the Informativity Principle will be emulated by Bayesianism.

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¹²In the case of human language (Goodman and Frank 2016), the same result can be conceptualized in two slightly different ways: (i) the addressee has access to the speaker’s strategy, and thus infers from the production of a less informative expression that the more informative one was false – as in the standard Gricean approach; (ii) the addressee has access to the speaker’s behavior. In the animal case, (i) might make little sense if no theory of mind is present, but (ii) makes good sense.

¹³One way to extend simple Bayesianism is to endow it with a strategic component. Signalers take into account the receiver’s behavior and realize that producing an eagle call is more helpful to them than producing a general call. This, of course, is the standard derivation of the Informativity Principle in human linguistics, in a long line of work pioneered by Grice and recently formalized in Bayesian pragmatics (e.g. Grice 1989; Horn 1972; Franke and Jäger 2016; Goodman and Frank 2016). But when applied to animals, this derivation may be implausible if they lack a theory of mind. This is the reason the Informativity Principle in animal linguistics was treated in attempts such as (1) as primitive, and without a commitment to a strategic component or a theory of mind.

¹⁴In that framework, strategies compete across different individuals rather than within the speaker’s mind, as in the human case (Franke and Jäger 2016; Goodman and Frank 2016). Populations using suboptimal strategies are in the end invaded by mutants using better strategies.

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